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IMMUNOPOTENTIATING EFFECT OF KHAMIRA MARWARID, AN HERBO-MINERAL PREPARATION

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SUMMARY

This study investigated the immunopotentiating effect of Khamira Marwarid (KM), an herbo-mineral Unani preparation, in an animal model. KM was administered to mice orally at a dose level of 2 g/kg body weight for 5, 10, 15 or 30 days, following which the animals were sacrificed for hematology and immune function testing, including lymphoid organ weight and cellularity. The group of animals receiving KM showed a significant increase ($P < 0.05$) in the relative organ (spleen and thymus) weight. Cellularity of the spleen and bone marrow also increased significantly ($P < 0.01$). Groups receiving KM for 10 and 15 days showed an increase in hemoglobin, red blood cells (RBCs) and total white blood cells (WBCs) ($P < 0.01$). The humoral immune response, evaluated by the plaque-forming cell (PFC) assay after challenging the mice with goat RBCs, was better in all treated groups when compared to controls. Maximum hemagglutination titer was obtained in mice treated with KM for 15 days. Ovalbumin-specific serum IgG levels in the treated mice also increased significantly ($P < 0.01$), suggesting an immunopotentiating effect for the preparation. Administration of KM resulted in elevated levels of IgG2a and IgG2b. A comparison of anti-ovalbumin IgE and IgG was also done; anti-ovalbumin IgE decreased, with a concomitant increase in anti-ovalbumin IgG. Administration of KM for 10 or 15 days elicited an increase in the delayed-type hypersensitivity (DTH) response. Taken together, the results suggest an immunostimulatory effect for KM through a mechanism leading to a Th1-dominant immune state.

Key words: Immunopotential - Traditional herbal preparation

INTRODUCTION

Immunosuppression is one of the major consequences of many diseases, including cancer and AIDS. In addition, radiotherapy and chemotherapy, used frequently for cancer treatment, also cause immunosuppression (1-3). Synthetic immunomodulating drugs such as levamisole, interferon and IL-2 in combination with cisplatin, adriamycin, 5-fluorouracil, etc., have been used for the treatment of cancer (4), but cause myalgias, fatigue, neutropenia, anorexia, elevated transaminases and/or proteinuria (5). The problem can be solved using immunomodulating drugs of herbo-mineral origin.

Numerous medicinal plants have been identified and developed as chemotherapeutic agents and as modulators of the immune system, and have been found to be able to enhance the function of B lymphocytes. Several of these plant products have been used widely throughout the globe and as starting materials for various prescriptions. A few examples include *Echinacea purpurea*, *Echinacea pallida*, *Panax ginseng*, *Eleutherococcus senticosus*, *Rehmania glutinosa* and *Brassica oleracea* (6-12). Compounds derived from *B. oleracea* have been found to stimulate the humoral immune response and enhance circulating antibody titers (12). Extracts from a number of plant products, such as *Withania somnifera*, *Tinospora cordifolia*, *Panax ginseng*, *Asparagus racemosus* and *Azadirachta indica*, also possess immunomodulating activity (13-17).

Khamira Marwarid (KM) is a Unani herbo-mineral preparation (18) from the Indian subcontinent that has been used as a rejuvenator

and vitalizer for centuries. It is prescribed as a tonic for the heart and the brain, usually as a recuperating agent after disease (19). The preparation is given especially to convalescent patients of typhoid and chickenpox to help them overcome fatigue and weakness after the disease (20), and is claimed to boost the body's capacity to fight infection and strengthen the vital organs. It is tempting to speculate that the restorative and recuperating power of this preparation may be due to its action on the immune system. We therefore evaluated the immunopotentiating effect of KM in a mouse model.

MATERIALS AND METHODS

Animals and chemicals

Swiss albino male mice, 6-8 weeks old and weighing 20-25 g, were used throughout the study. Mice were kept in an environmentally controlled room in polypropylene cages in groups of six each in the institute's Central Animal House Facility. They were fed a commercial pellet diet and water ad libitum. The study was approved by the Animal Ethics Committee and conformed to the national guidelines on the care and use of laboratory animals. Hanks' balanced salt solution (HBSS), ovalbumin, guinea pig complement, o-phenylene diamine dihydrochloride (OPD) and immunochemicals were purchased from Sigma-Aldrich Co. (USA). All other chemicals were of analytical grade and procured from the standard commercial sources in India.

Preparation of KM

KM is a product of Hamdard Wakf Laboratories (New Delhi, India). It is a polypharmaceutical preparation containing crude drugs of plant and mineral origin in the form of a fine powder or their infusion/ decoction in sugar syrup. KM is prepared by adding microfine powder of minerals such as pearl, Yashab and Zahar Mohra to the sugar syrup of a specific consistency. Crude extracts of *Punica granatum*, *Pyrus malus*, Sandal Safed, Kehruba and Warq-e-Nuqurah are then added to the cooled mixture with continuous stirring. The mixture is blended until the required consistency is achieved. The composition of the prescription is shown in Table I.

Dosage and treatment protocol

The medicine was suspended in normal saline and administered orally to mice with the help of a cannula at a dose of 2 g/kg body weight in a volume not exceeding 0.2 mL per animal. The dose was based on the human equivalent. Mice in the control group received a similar volume of normal saline. Mice were divided into eight groups (I-VIII), each consisting of six mice. KM was administered orally once a day for the specified duration of time. Groups I-IV were administered KM for 5, 10, 15 and 30 days, respectively. The mice in the respective control groups (V-VIII) received normal saline.

Lymphoid organ weight, cellularity and body weight

Body weight gain and relative organ weight (organ weight/100 g of body weight) of kidney, liver, spleen and thymus were determined for each mouse. A single cell suspension was prepared in HBSS from bone marrow, spleen and thymus for cell count. Body weights of all mice were recorded prior to sacrifice, i.e., 24 h after administering the last dose of KM.

Table I. The composition of Khamira Marwarid.

S. No.	Name	Origin	Major chemical composition	Content (g/10 g)
1.	Pearl	Mineral	Calcite or calcium carbonate	0.24
2.	Jade	Mineral	Jadite (AlSi ₂ O ₆) and nephrite [Ca ₂ (MgFe) ₅ (OH) ₂ (SiO ₄) ₂]	0.12
3.	Serpentine	Mineral	Bowerite, lizardite and chrysotile	0.12
4.	Argentum	Mineral	Silver	0.12
5.	<i>Santalum album</i> Linn.	Plant	Santalol	0.12
6.	<i>Punica granatum</i> Linn.	Plant	Punicalagin	0.011
7.	<i>Pyrus malus</i> Linn.	Plant	Quercetin	0.011
8.	<i>Vateria indica</i> Linn.	Plant (resin)	Fatty acids C16, C18, C18:1, C18:2, C20	0.12

Hematological parameters

Blood was collected from the treated and the control group of mice 24 h after the last dose. Total white blood cell (WBC) count, differential leukocyte count, red blood cell (RBC) count and hemoglobin content were measured using an automated hematology cell counter (Sysnex K-1000).

Assessment of humoral immune function

For assessing humoral immune function, mice from various groups were immunized with 0.1 mL goat RBC suspension comprising 325 x 10⁶ cells in the tail vein 24 h after administering the drug. On the fifth day after the antigen was administered, the hemagglutination titer assay (HA) and the plaque-forming cell assay (PFC) were performed.

The HA was performed using the reported protocol (21). Briefly, blood was collected from the retro-orbital plexus of each animal and the serum was separated by centrifugation technique. A twofold serial dilution of serum was added to the wells in the microtiter plate already coated with 50 µL of 1% goat RBC suspension in PBS. After mixing, the plate was incubated at room temperature for 2 h. The antibody titer was expressed as a reciprocal of the highest anti-sera dilution showing visible hemagglutination.

The PFC assay was performed according to the standard procedure. Animals were sacrificed on the fifth day and the spleen was removed, and a single cell suspension of 1 x 10⁶ cells/mL was prepared in HBSS. For the PFC assay, the goat RBCs were prepared at a cell density of 96 x 10⁶ cells/mL and guinea pig complement was diluted 1:10 with normal saline. The assay mixture was prepared by mixing splenocytes, goat RBCs and complement at a ratio of 2:2:1 and incubated at 37 °C for about 2.5 h. The number of plaques developed was counted microscopically and expressed as PFCs/10⁶ spleen cells.

Estimation of ovalbumin-specific antibody level and titer in serum using ELISA

For this analysis, mice were injected intramuscularly with 50 µg ovalbumin emulsified with Freund’s complete adjuvant (CFA) after the completion of drug administration. A booster dose (25 mg of ovalbumin in CFA) was given on days 21 and 35. Mice were bled from the retro-orbital vein on days 27, 42 and 60, and the serum was separated and evaluated for ovalbumin-specific IgG levels and IgG peak titers by the ELISA method. Briefly, wells of the ELISA plate (Griener, Germany) were coated with 100 mL of ovalbumin (100 ng/well) in 50 mM carbonate–bicarbonate buffer (pH 9.6) and incubated at 4 °C for 12 h. Following incubation, wells were washed three times with PBS containing 0.05% Tween-20. The free binding sites were blocked by adding 300 µL of PBS containing 3% skimmed milk powder to each well, and incubating at 37 °C for 12 h. The wells were washed again and 100 mL of serum at a dilution of 1:400 in PBS was added to each well. The mixture was incubated further for 1 h at 37 °C, and then the wells were washed, followed by the addition of 100 mL of horseradish peroxidase (HRPO)-conjugated goat anti-mouse IgG (Sigma-Aldrich Co., USA) diluted 1:1000 in PBS. The mixture was incubated for 1 h at 37 °C. Wells were washed again, and 100 mL of substrate (containing 4 mg of OPD and 10 mL of H₂O₂ in 10 mL of 10 mM citrate–phosphate buffer at pH 5.0) was added to

each well. After developing the color, the reaction was terminated by adding 50 mL of 4 M H₂SO₄. Absorbance was measured at 492 nm in an automatic microplate reader (ECIL, Hyderabad). Titers of anti-ovalbumin antibody of IgG isotype were determined using the same procedure, except that serial dilutions of pooled sera of each group were used.

Estimation of ovalbumin-specific serum anti-isotypic antibodies using ELISA

The type of IgG subclass elicited was elucidated using a Sigma iso-typing kit. For ELISA, the same procedure as described above was used until the addition of antisera. The wells were then washed and 100 mL of goat anti-mouse IgG isotypes (IgG1, IgG2a, IgG2b and IgG3) (Sigma, St. Louis, MO, USA) diluted 1:1000 in PBS were added to each well and incubated for 2 h at 37 °C. After the usual washing steps, rabbit anti-goat IgG-HRPO conjugate was added and incubated for 1 h at 37 °C. After the usual washing step, color was developed as above, and the optical density was measured at 492 nm in an automatic microplate reader (ECIL, Hyderabad).

Comparative ovalbumin-specific serum IgE and IgG levels

Comparative levels of IgE and IgG were evaluated using ELISA. The same protocol as mentioned above was followed until the addition of ovalbumin antisera at a dilution of 1:200 for 2 h at 37 °C. Subsequently, the plates were washed three times with PBS-Tween-20 and a second antibody, goat anti-mouse IgE HRPO and goat anti-mouse IgG HRPO, was added simultaneously to the same antisera in adjacent wells. After the usual washing step, color was developed using H₂O₂ as substrate and OPD as chromogen. The reaction was stopped with 4 M H₂SO₄ and quantified with the ELISA reader at 492 nm.

Delayed-type of hypersensitivity (DTH) response

The DTH response was determined as described earlier (22). Briefly, after completing the treatment, animals were immunized with 1 × 10⁹ goat RBCs subcutaneously. After 5 days, all animals were again

Table II. Effect of Khamira Marwarid on relative organ weights in treated mice.

Group	Relative organ weight (g/100 g body weight) (mean ± S.E.)			
	Spleen	Thymus	Liver	Kidney
C5	0.82 ± 0.1	0.24 ± 0.006	6.72 ± 0.44	0.99 ± 0.03
A5	0.84 ± 0.09	0.24 ± 0.01	6.92 ± 0.58	0.99 ± 0.16
C10	0.82 ± 0.06	0.23 ± 0.01	6.68 ± 0.58	0.92 ± 0.07
A10	0.88 ± 0.09	0.27 ± 0.01 ^a	6.79 ± 0.15	0.94 ± 0.02
C15	0.84 ± 0.05	0.24 ± 0.01	6.78 ± 0.84	1.02 ± 0.10
A15	1.11 ± 0.04 ^a	0.29 ± 0.01 ^a	6.88 ± 0.63	0.91 ± 0.13
C30	0.85 ± 0.05	0.24 ± 0.02	6.69 ± 0.21	1.10 ± 0.10
A30	0.92 ± 0.05	0.26 ± 0.01	6.63 ± 0.13	0.96 ± 0.03

Animals were sacrificed 24 h after the last dose of the preparation. Organ weights were recorded and calculated as g/100 g body weight. Data represent the mean ± S.E. (n = 6). A, treated with KM; C, control. ^aP < 0.05 when compared to the respective control.

Table III. Effect of Khamira Marwarid on cellularity of lymphoid organs.

Group	Cellularity (mean ± S.E. × 10 ⁶)		
	Bone marrow	Thymus	Spleen
C5	15.46 ± 0.76	58.25 ± 8.36	320.71 ± 12.70
A5	20.22 ± 1.42 ^b	66.62 ± 8.08	374.29 ± 16.8
C10	15.96 ± 1.35	59.68 ± 4.2	310.96 ± 19.21
A10	24.80 ± 1.03 ^c	68.81 ± 8.11	398.71 ± 13.30 ^b
C15	16.03 ± 0.76	56.14 ± 7.39	316.26 ± 20.78
A15	29.01 ± 1.61 ^c	70.76 ± 5.31	410.33 ± 10.65 ^b
C30	15.53 ± 0.24	54.96 ± 4.01	310.05 ± 24.24
A30	32.01 ± 0.57 ^c	75.15 ± 4.3 ^b	386.03 ± 9.8 ^a

Animals were sacrificed 24 h after the last dose of the preparation. Bone marrow cells were collected from femoral bone. A single cell suspension of bone marrow, thymus and spleen was prepared in HBSS. Cells were counted using a Neubauer chamber. Data represent mean ± S.E. (n = 6). A, treated with KM; C, control. ^aP < 0.05 when compared to the respective control; ^bP < 0.01 when compared to the respective control; ^cP < 0.001 when compared to the respective control.

challenged with 1 × 10⁸ goat RBCs in the left hind footpad. The right footpad, injected with the same volume of normal saline, served as control. The increase in the footpad thickness was measured 24 h after the last challenge with goat RBCs using dial calipers.

Statistical analysis

The statistical significance of the differences between treated and control groups was determined using the Student's *t*-test; *P* < 0.05 was chosen as the level of significance.

RESULTS

Effect of KM on the body weight, lymphoid organ weight, cellularity and toxicity

Mice administered KM did not show any significant gain in body weight when compared with the untreated control group. However, a significant increase (*P* < 0.05) in the weight of the spleen was found in mice receiving the drug for 15 days (Table II). The size and weight of the thymus also increased significantly (*P* < 0.05). No significant change in the weights of the kidney and liver was observed in animals treated with KM.

A significant increase (*P* < 0.01) in bone marrow cellularity was also a prominent feature. Bone marrow cellularity increased even after 5 days of herbal treatment. The cellularity of the spleen was also significantly increased (*P* < 0.01) after 10 and 15 days of treatment with the herbal preparation. The cellularity of the thymus increased (*P* < 0.01) in mice after 30 days of treatment (Table III). Oral administration of KM for a period of 30 days did not produce mortality or show any signs of toxicity.

Effect of feeding KM on hematological parameters

Administration of KM for 15 days resulted in a significant increase (*P* < 0.001) in the hemoglobin content and RBC count of normal

Table IV. Effect of Khamira Marwarid on blood cells.

Group	Hemoglobin concentration (g%)	RBC count in million/cmm	WBC cpunt in thousand/cmm
C5	11.60 ± 0.55	7.11 ± 0.33	10.00 ± 0.71
A5	13.16 ± 0.46	8.16 ± 0.43	12.10 ± 0.75
C10	11.26 ± 0.88	7.10 ± 0.15	11.26 ± 0.95
A10	14.96 ± 0.25 ^b	8.93 ± 0.40 ^b	16.20 ± 0.66 ^b
C15	12.11 ± 0.86	7.4 ± 0.16	11.88 ± 0.92
A15	16.96 ± 0.40 ^c	9.03 ± 0.17 ^c	17.83 ± 0.85 ^b
C30	14.9 ± 0.26	7.8 ± 0.61	12.20 ± 0.83
A30	17.66 ± 0.28 ^c	9.89±0.17 ^c	17.63±0.62 ^c

Blood was collected 24 h after the last dose of the preparation. Data are represented as mean ± S.E. (n = 5). A, treated with KM; C, control. ^b*P* < 0.01 when compared to the respective control; ^c*P* < 0.001 when compared to the respective control.

Swiss albino mice treated for 30 days. The total WBC count increased in all treated groups. Animals receiving KM for 15 days showed about a 50% increase in the total leukocyte count (Table IV). However, the lymphocyte to neutrophil ratio did not show any significant difference (data not shown).

Effect of KM on humoral immunity

The PFC (Fig. 1) and hemagglutination assays (Table V) were performed to study the humoral response. The number of antibody-forming cells in the spleen was significantly increased (*P* < 0.001)

Table V. Effect of Khamira Marwarid on antibody titer.

Groups	Hemagglutination titer
C5	1:128
A5	1:384
C10	1:128
A10	1:1024
C15	1:128
A15	1:4096
C30	1:128
A30	1:768

After the final dose of the preparation, mice were immunized with 1 x 10⁹ goat red blood cells (RBCs) i.v. Sera were collected on the 5th day of immunization. The hemagglutination assay was performed on pooled sera of each group. The hemagglutination titer represents the reciprocal of the highest dilution that showed visible agglutination. A, treated with KM; C, control.

(Fig. 1). Even a 5-day treatment with KM before the antigen was injected resulted in an increase in the number of antibody-forming cells. The maximum increase in PFCs (273 PFC/10⁶ splenocytes) was observed in the group that received KM for 15 days when compared to the control, where the number of PFCs was 122 PFC/10⁶ splenocytes. In all groups, the number of antibody-forming cells increased significantly.

The amount of circulating anti-goat RBC antibodies was greater in the sera of treated mice compared to the untreated control group, as indicated by higher hemagglutination titers (Table V). The maximum

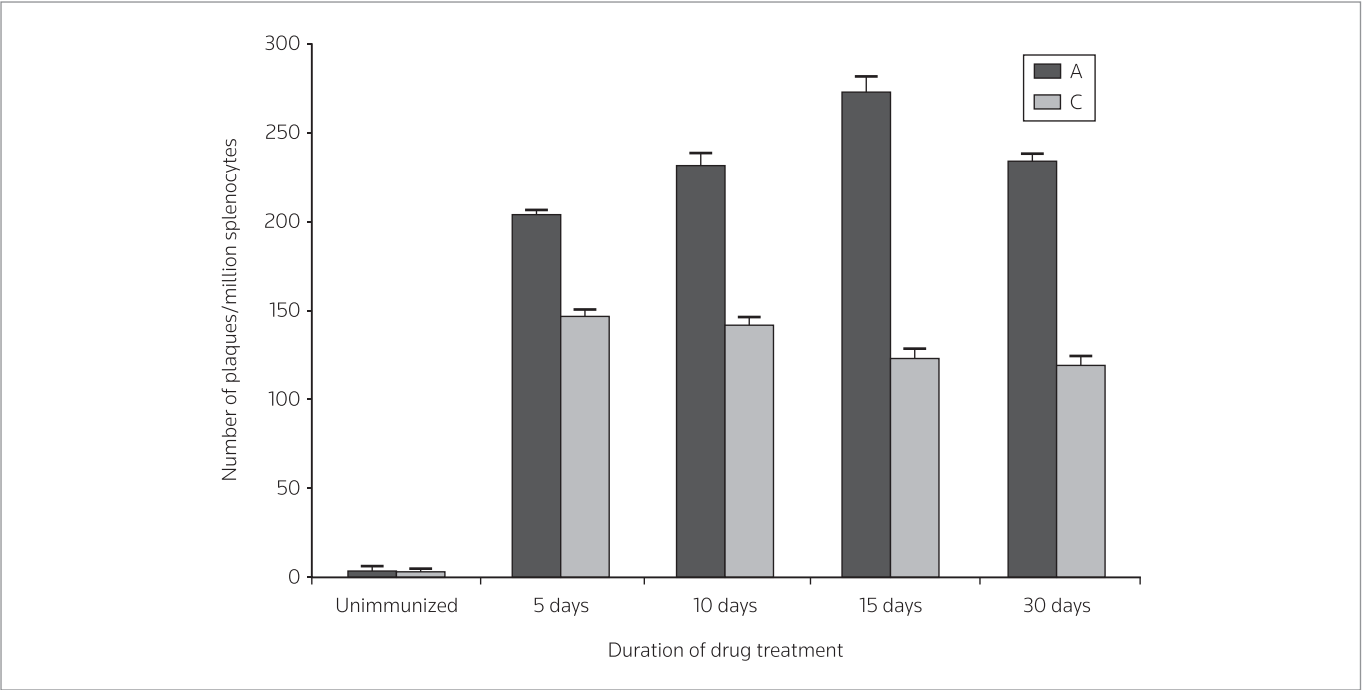


Figure 1. Effect of Khamira Marwarid (KM) on number of antibody-forming cells as measured by the plaque-forming cell (PFC) assay. Mice were administered KM orally for different time intervals and then immunized with goat red blood cells (RBCs). Spleen cells were isolated on the fifth day after immunization with goat RBCs and the PFC assay was performed in liquid medium. Data are represented as mean + S.E. (n = 5). A, treated with KM; C, control.**P* < 0.01 when compared to the respective control.

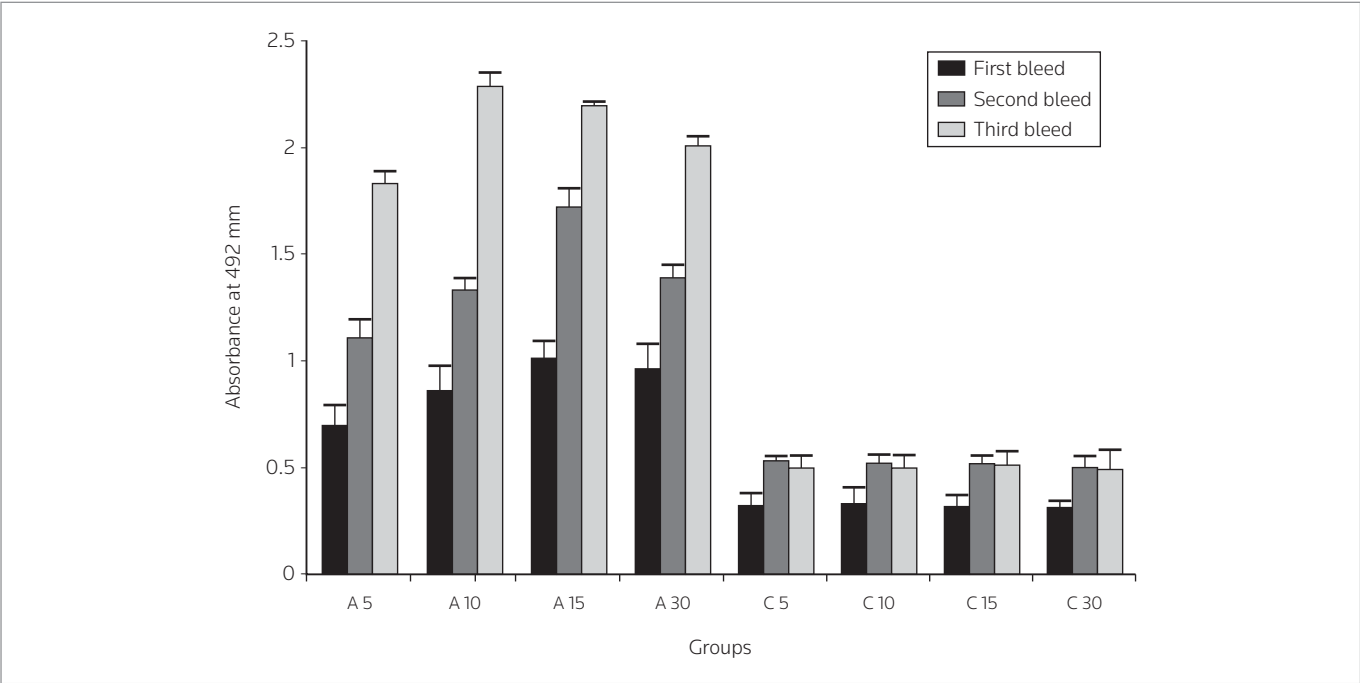


Figure 2. Effect of Khamira Marwarid (KM) on ovalbumin-specific IgG levels. After the final dose of the preparation, animals were immunized with ovalbumin (50 µg/mouse) and booster doses were given on days 21 and 35. Sera were collected on days 27, 42 and 60 after the first immunization and the ovalbumin-specific IgG antibody response was measured by ELISA. Data are represented as means ± S.E. (n = 5). A, treated with KM; C, control. *P < 0.01 when compared to the respective control.

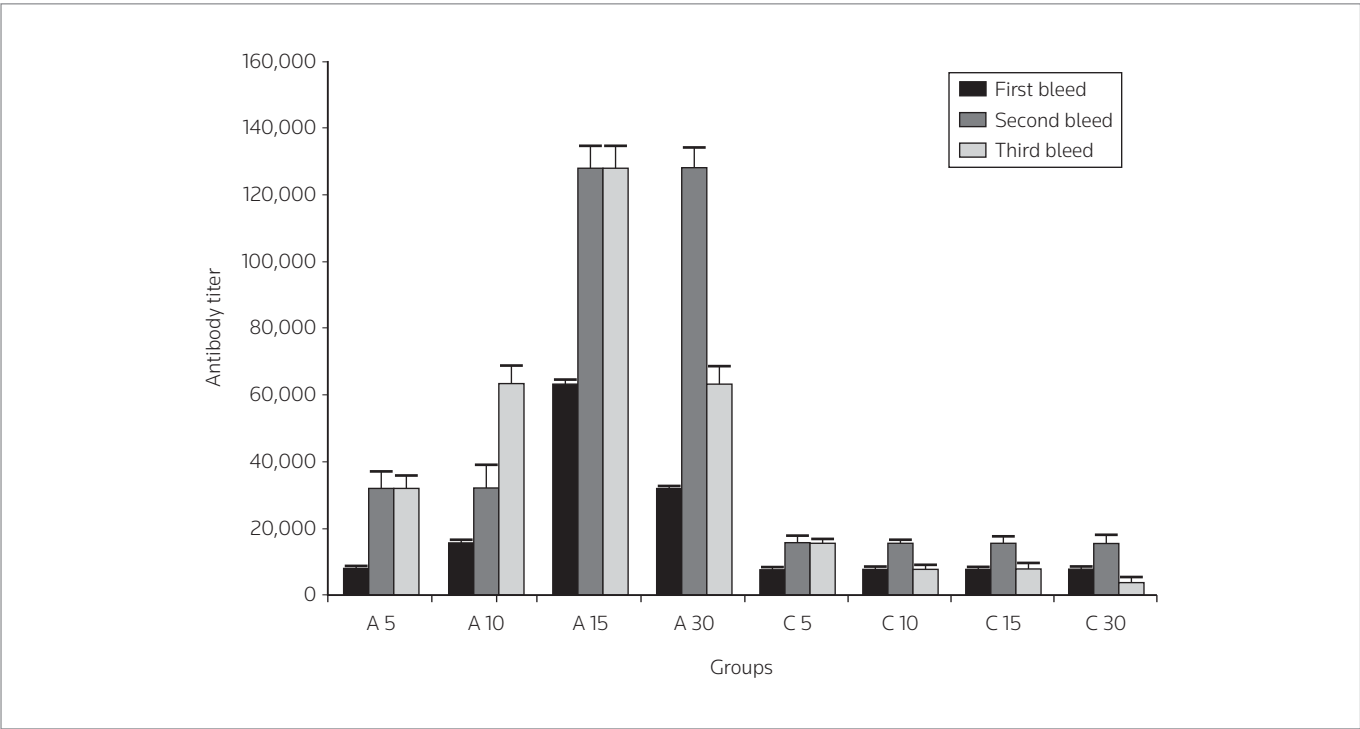


Figure 3. Effect of Khamira Marwarid (KM) on ovalbumin-specific IgG titers. Serial dilutions of the pooled sera were done and titers of ovalbumin-specific IgG isotypes were measured using ELISA. Data are represented as means ± S.E. (n = 5). A, treated with KM; C, control. *P < 0.01 when compared to the respective control.

difference in hemagglutination titer was observed in the group receiving KM for 15 days.

Effect on ovalbumin-specific serum IgG levels and IgG titer

Oral administration of KM enhanced the induction of ovalbumin-specific IgG antibody responses in mice, as determined by ELISA (Fig. 2). A significant increase in the total ovalbumin-specific IgG level was observed in mice given KM for 5, 10, 15 or 30 days. The maximum increase in ovalbumin-specific IgG level was observed in animals receiving the drug for 10 and 15 days. Also, the secondary immune response seen in the second bleed was statistically higher ($P < 0.01$) compared to that of the first bleed in all treated groups. Interestingly, the high ovalbumin-specific IgG levels were maintained in the third bleed, that is 60 days after primary immunization.

The antibody titer as determined by ELISA in the pooled serum also increased in mice administered the preparation for different time intervals (Fig. 3). The maximum endpoint titer value peaked to 128,000 in the second bleed of mice treated for 15 and 30 days.

Effect of KM on the levels of ovalbumin-specific IgG subtypes

The IgG subclasses were estimated to get an idea of the quality of the immune response generated, as well as to delineate the Th1–Th2 dichotomy, which is known to play a crucial role in the development or containment of the disease. Oral administration of KM resulted in the generation of all four IgG isotypes (Fig. 4), albeit at very low levels. Predosing of mice with KM resulted in marked upregulation of ovalbumin-specific IgG2a and IgG2b for all the treated groups as compared to controls. Mice predosed with KM for 5 days showed an

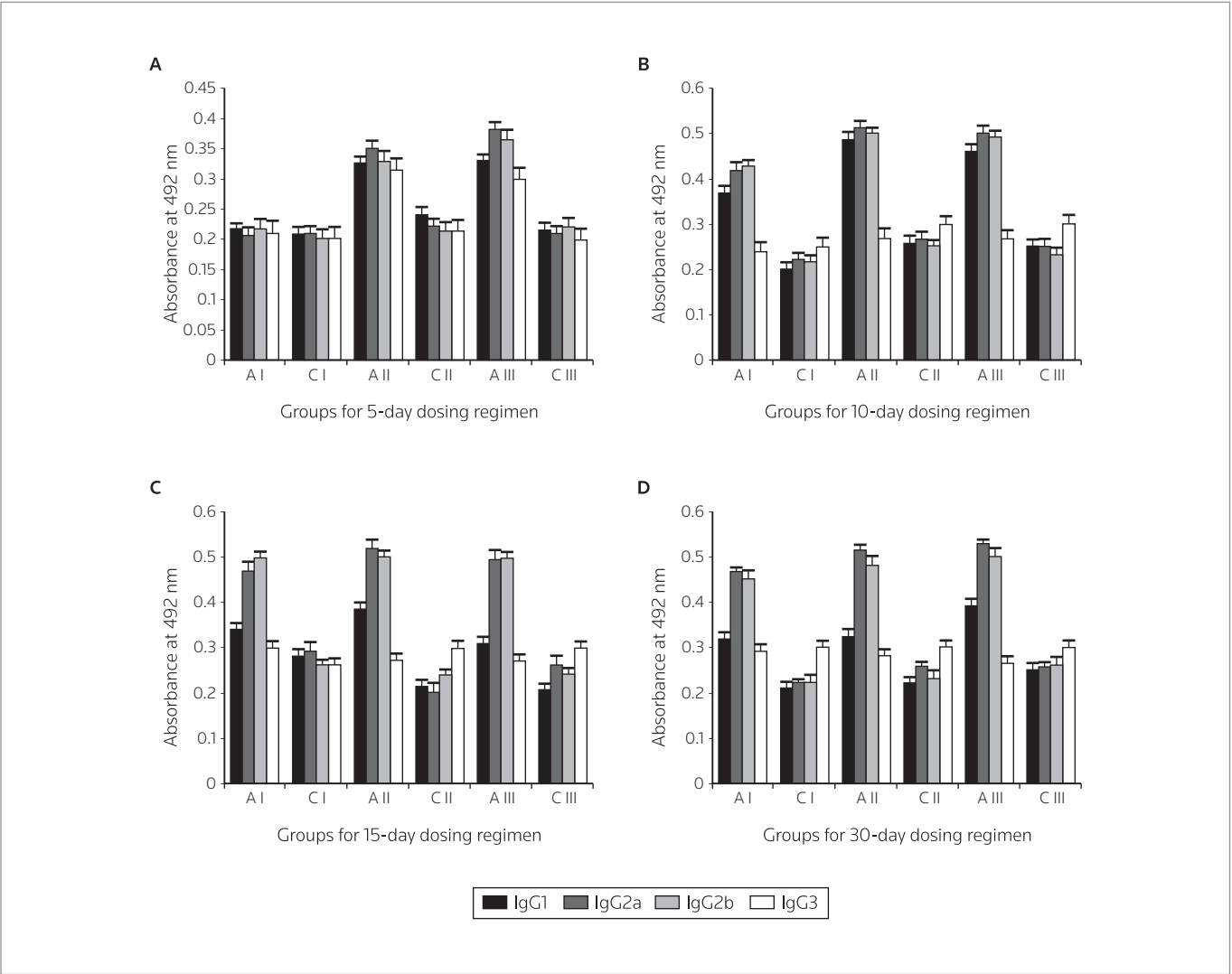


Figure 4. Effect of Khamira Marwarid (KM) on ovalbumin-specific IgG subtypes. After the final dose of the preparation, animals were immunized with ovalbumin (50 µg/mouse) and booster doses were given on days 21, 35 and 60 after immunization and ovalbumin-specific IgG subtypes were measured using an IgG isotyping kit. Data are represented as means ± S.E. (n = 5). A, treated with KM; C, control. * $P < 0.01$ when compared to the respective control.

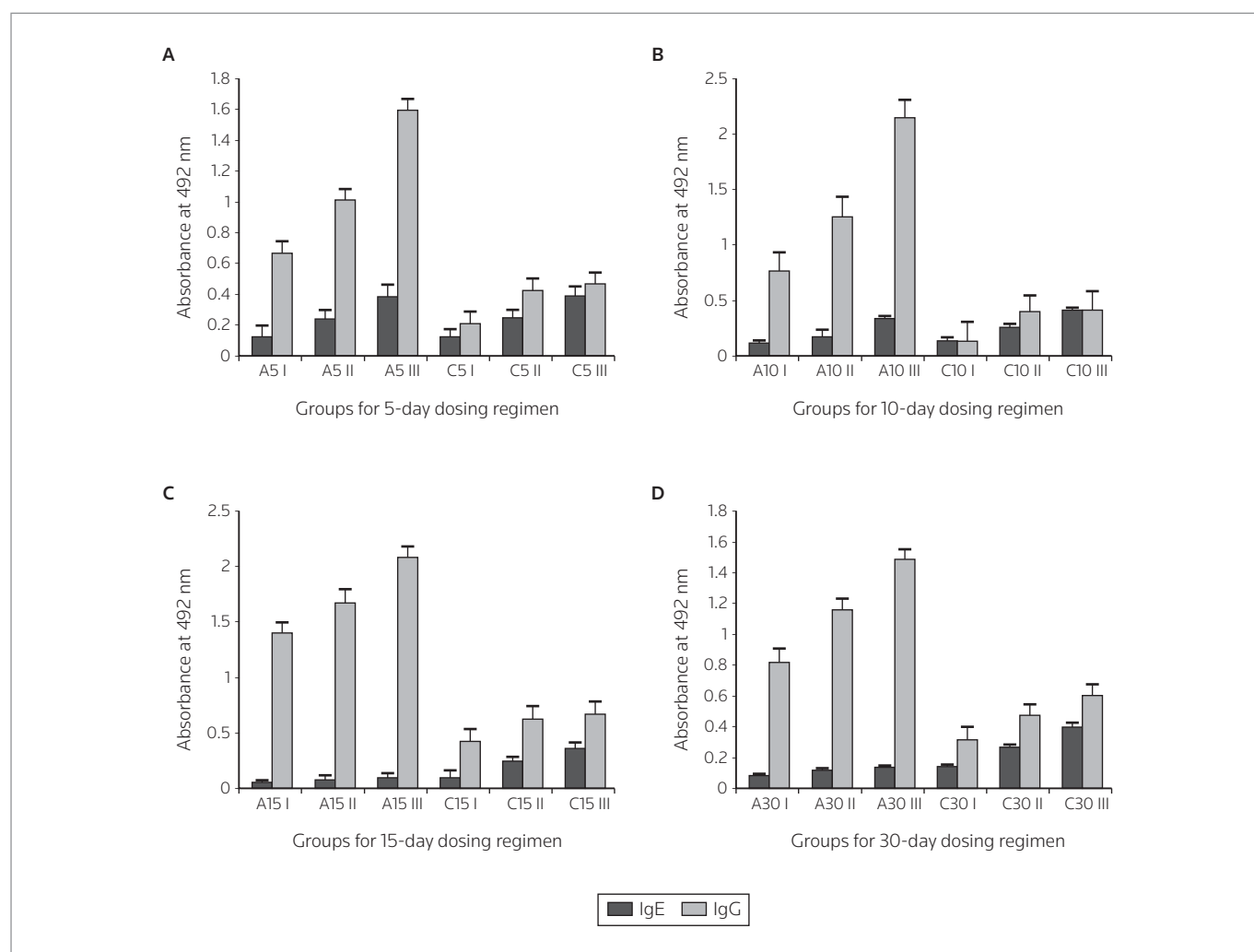


Figure 5. Effect of Khamira Marwarid (KM) on comparative ovalbumin-specific IgE and IgG levels. After the final dose of the preparation, animals were immunized with ovalbumin (50 µg/mouse) and booster doses were given on days 21, 35 and 60 after immunization. Ovalbumin-specific IgE and IgG levels were measured simultaneously using ELISA. Data are represented as means ± S.E. (n = 5). A, treated with KM; C, control. * $P < 0.01$ when compared to the respective control.

increase in the induction of all four subtypes in the second bleed. However, a strong reduction in the level of IgG3 was observed in mice pretreated with KM for 10 days, which was maintained even in the 30 days predosing group. Induction of levels of isotype IgG1 were suppressed in mice after 15 days of KM administration compared to mice treated for 5 and 10 days.

Effect of KM on comparative levels of IgE and IgG

Immunization of mice with ovalbumin is known to exhibit upregulation of Th2 cytokine and serum anti-ovalbumin IgE levels. The effect of KM administration on comparative IgE and IgG production in vivo was examined to analyze any shift, if present, in the levels of IgE to IgG, or vice versa. Immunization with ovalbumin resulted in upregulation of both IgE and IgG in the second and third bleed in groups of mice pretreated with KM for 5 or 10 days (Fig. 5). However, levels of

anti-ovalbumin IgE decreased in mice predosed with KM for 15 days as compared to control mice receiving normal saline instead of KM. This decrease in the level of IgE was accompanied by an upregulation in the level of IgG, which was maintained until the last bleed at 30 days in the predosed group.

Effect of KM on cell-mediated immunity

To assess the effect of oral administration of KM to mice on the cell-mediated immune response, the DTH response to immunization with antigen was studied. No significant increase ($P < 0.01$) in the footpad thickness in the unimmunized mice treated with KM was found when the group was compared to the unimmunized mice treated with normal saline (Fig. 6). A significant increase in the footpad thickness was observed in the goat RBC-immunized mice. The DTH response was further significantly enhanced ($P < 0.01$) in goat

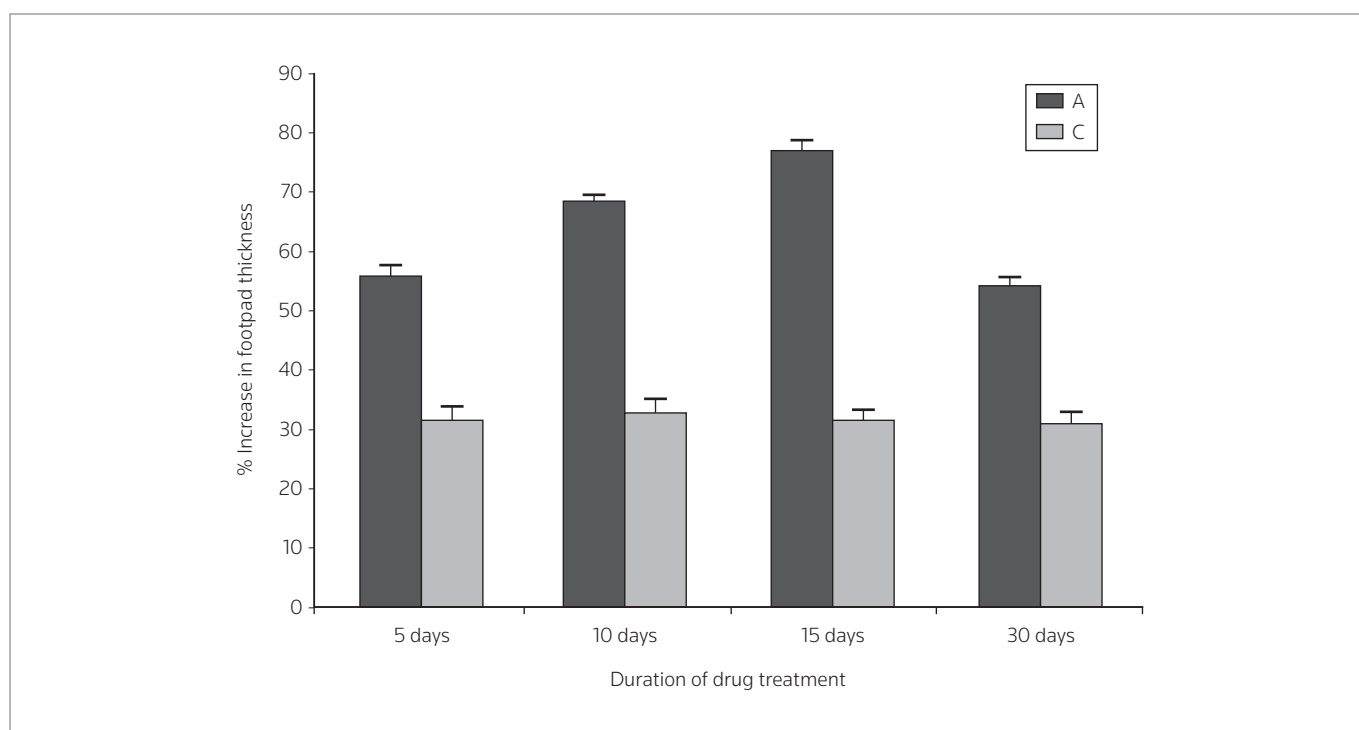


Figure 6. Effect of Khamira Marwarid (KM) on the delayed-type hypersensitivity (DHT) response. After the final dose of the preparation, mice were immunized with goat red blood cells (RBCs) subcutaneously. On day 5, mice were again challenged with goat RBCs in the left hind footpad; normal saline injected into the right footpad served as trauma control. The increase in footpad thickness was measured 24 h after the last challenge. Data are represented as means $\pm$ S.E. ($n = 5$). A, treated with KM; C, control. * $P < 0.05$ when compared to the respective control.

RBC-immunized mice treated with KM. A significant increase in the DTH response was observed in mice treated with KM for 10 and 15 days prior to immunization with goat RBC. The increase in the footpad thickness of mice treated for 10 and 15 days was 68% and 76.88%, respectively; in the control group, the increase was 37.56% and 31.0%, respectively. However, prolonged administration of KM did not produce a statistically significant increase.

DISCUSSION

Administration of KM for 15 days at a dose of 2 g/kg body weight improved the overall health of mice, as indicated by an increase in hemoglobin content and RBC count, and strengthened the specific immune system. The maximum effect was observed when the preparation was administered to mice for 15 days continuously. Administration of KM for 10 or 15 days increased the bone marrow cell count, indicating its effect on hematopoietic stem cells of bone marrow. Also, the enhanced weight of the thymus and spleen, which represent primary and secondary organs of the specific immune system, suggested an immunopotentiating effect for KM. The increase in thymus weight was accompanied by an increase in cell count, which might have been partly due to the stimulating effect of KM on the maturation of lymphocytes and bone marrow hematopoietic cells that ultimately home in the thymus. However, the homing may be temporary, and in due course, normalization may ensue.

KM produced a significant increase in total WBC count as compared to control. The increase was more in the group treated for 15 days (Table IV). This is possible only when the hematopoietic system is stimulated.

The increase in PFCs and the level of anti-goat RBC antibody in the serum of mice receiving KM prior to immunization with goat RBC as compared to controls (Fig. 1, Table V) indicate stimulation of the specific immune system by the preparation.

KM enhanced humoral immunity, as determined by the level of anti-ovalbumin antibody of IgG type, which increased significantly in treated mice (Fig. 2). High levels of IgG antibody against ovalbumin were maintained until day 60 after immunization, suggesting that the drug had a prolonged effect on the potentiation of the humoral immune response. The augmentation of the humoral immune response to ovalbumin and goat RBCs by KM may be due to enhanced responsiveness of macrophages and T- and B-lymphocyte subsets involved in antibody synthesis (23). Since macrophages play an important role in coordinating the processing and presentation of antigen to T and B cells, KM is suggested to act by facilitating these processes.

Oral administration of KM was found to influence the Ig-specific isotype profile. A comparison of isotypes produced by mice to immunization with ovalbumin between treated and control animals showed the induction of two major IgG subclasses, IgG2a and IgG2b. This type of isotype induction profile is similar to that observed in natural immunity arising from a viral or bacterial infection. Viral infection in mice

induces an IgG response in which IgG2a accounts for 65-92% of total specific antibody. IgG2a has also been shown to protect against protozoal infections (24). IgG2a antibodies can mediate both C fixation and antibody-dependent cellular cytotoxicity in mice (25).

The regulation of IgE production is of clinical importance since this antibody plays a major role in the type I allergic response. In this study we demonstrate, using an in vivo murine model, that KM causes a reduction in ovalbumin-specific serum IgE. Moreover, a strong upregulation of the ovalbumin-specific IgG response was observed. Interestingly, these results, in terms of increase in IgG and other subclasses and suppression of the ovalbumin-specific IgE antibody response, may be comparable to those for *Quillaja saponaria* (Molinaceae) (26). This reduction of ovalbumin-specific IgE and IgG1, in addition to the enhancement of IgG2a, suggests that KM modulates the Th1/Th2 balance in favor of a Th1-type response.

A significant increase in the paw thickness in KM-treated mice was observed when compared to the mice in the control group ($P < 0.05$). T_{eff} cells, the effector cells that are responsible for the DTH reaction, recruit other cells (such as macrophages) and activate them (27), thereby enhancing the T- and B-cell responses (28). Treatment with KM probably caused a significant activation of macrophages, resulting in increased infiltration of polymorphonuclear cells. Further investigations on the effect of oral administration of KM on macrophages and lymphocytes need to be done to confirm the effect. It is evident from the present study that KM enhances both humoral and cell-mediated immune responses, which might be responsible for the beneficial effect of the preparation.

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DISCLOSURE

The authors have nothing to disclose.

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